BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Iteration: WHILE Loops

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. When is a count-controlled loop suitable?

2. Why use pseudocode?
   A) Run directly on the CPU
   B) Plan steps without exact syntax
   C) Replace all program testing
   D) Draw the user interface

3. "A diamond is used only for unconditional calculations." Correct this. Explain why.

4. A program works out a bill, then applies a discount. "The order of steps never matters." Correct this. What could go wrong?

Today's objectives

Write condition-controlled loops. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

WHILE loop
Checks a condition before each iteration; repeats while it is true.

Condition
A comparison such as attempts < 3 controls continuation.

Update
Change loop variables so the condition can eventually become false.

Unknown repetition
WHILE suits a task where the number of tries is not fixed beforehand.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

WHILE loop
Checks a condition before each iteration; repeats while it is true. For example, consider a password attempt counter.

Condition
A comparison such as attempts < 3 controls continuation. For example, consider a quiz retry.

Update
Change loop variables so the condition can eventually become false. For example, consider a low stock checker.

Unknown repetition
WHILE suits a task where the number of tries is not fixed beforehand. For example, consider a temperature alarm.

I Do - Worked model

A password attempt counter is being developed. Explain how while loop and condition are used. [4 marks]

- Identify where this particular program needs control or values.
- WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1].
- Condition [1]; A comparison such as attempts < 3 controls continuation. [1].
- Give one accurate point and a linked reason for each technique.

We Do - Guided practice

1. A quiz retry is being programmed. Explain why while loop matters. [2 marks]

2. For a low stock checker, explain how condition is used. [2 marks]

3. A temperature alarm needs update and unknown repetition. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Add one line so the loop reads another answer and can finish.

```
answer = input("Continue? Y/N")
while answer != "N"
...
endwhile
print("Stopped")
```

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how condition and update can help a correction prompt. [4 marks]

Q4. Explain the roles of while loop, update and unknown repetition when creating a survey response. [6 marks]

Exit Ticket - True or False?

Checks a condition before each iteration; repeats while it is true.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. When the number of repetitions is known or determined by a fixed count.

2. B - Plan steps without exact syntax

3. A decision or condition with alternative paths.

4. Steps must be clear and in the right order. Check that conditions give the right result for every allowed input.

I Do - Explain the model

Identify where this particular program needs control or values.

WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1].

Condition [1]; A comparison such as attempts < 3 controls continuation. [1].

Give one accurate point and a linked reason for each technique.

We Do - Model answers

1. WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1].

2. Condition [1]; A comparison such as attempts < 3 controls continuation. [1].

3. Update [1]; Change loop variables so the condition can eventually become false. [1]. Unknown repetition [1]; WHILE suits a task where the number of tries is not fixed beforehand. [1].

Code First - Answer

```
answer = input("Continue? Y/N")
while answer != "N"
answer = input("Continue? Y/N")
endwhile
print("Stopped")
```

Expected output: Stopped (test input sequence: Y, then N)

You Do - Self-mark

Q2: WHILE loop [1]; Update [1].

Q3: Condition [1]; A comparison such as attempts < 3 controls continuation. [1]. Update [1]; Change loop variables so the condition can eventually become false. [1].

Q4: WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1]. Update [1]; Change loop variables so the condition can eventually become false. [1]. Unknown repetition [1]; WHILE suits a task where the number of tries is not fixed beforehand. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.